Chinese Remainder Theorem (CRT)

$x \equiv a_1 \pmod{P_1}$

$x \equiv a_2 \pmod{P_2}$

$x \equiv a_3 \pmod{P_3}$

Formula:

$\therefore x \equiv a_1 M_1 Y_1 + a_2 M_2 Y_2 + \dots + a_n M_n Y_n \pmod{M_n}$

$m = P_1 \cdot P_2 \cdot P_3$ | $M_n = \frac{m}{P_n}$

$x \equiv 2 \pmod{3} \rightarrow P_1$

$x \equiv 3 \pmod{5} \rightarrow P_2$

$x \equiv 2 \pmod{7} \rightarrow P_3$

$\boxed{a_1 = 2,\ a_2 = 3,\ a_3 = 2}$

$m = 3 \cdot 5 \cdot 7$
$= 105$

$M_1 = \frac{105}{3} = 35$

$M_2 = \frac{105}{5} = 21$

$M_3 = \frac{105}{7} = 15$

$M_n Y_n \equiv 1 \pmod{P_n}$

Y_n = M_n কে এমন নম্বর এর সঙ্গে গুণ দিতে হবে, যাকে সেই নম্বর P_n এর সঙ্গে mod করলে ভাগশেষ এক হয়

$\therefore 35 Y_1 \equiv 1 \pmod{3}$

$\Rightarrow 35 \cdot 2 \equiv 1 \pmod{3}$

$\Rightarrow \boxed{Y_1 = 2}$

$\therefore 21 \cdot Y_2 \equiv 1 \pmod{5}$

$\Rightarrow 21 \cdot 1 \equiv 1 \pmod{5}$

$\Rightarrow \boxed{Y_2 = 1}$

$\therefore 15 \cdot Y_3 \equiv 1 \pmod{7}$

$\Rightarrow 15 \cdot 1 \equiv 1 \pmod{7}$

$\Rightarrow \boxed{Y_3 = 1}$

$\therefore x = a_1 M_1 Y_1 + a_2 M_2 Y_2 + a_3 M_3 Y_3 \pmod{M_n}$

$= 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 \pmod{105}$

$= 233 \pmod{105}$

$\therefore 233 / 105 = 2.219\dots$

$\Rightarrow 233 - (105 \times 2)$

$= 233 - 210$

$= 23$ (Ans)

Trial Division

1 এর বড় & not a prime.

⇒ If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

∴ Suppose, n is composite,

∵ Then, $n = a \times b$

∴ Using proof by contradiction, suppose both factors are greater than $\sqrt{n}$

⇒ $a > \sqrt{n}$, $b > \sqrt{n}$

⇒ $a \cdot b > \sqrt{n} \times \sqrt{n}$

⇒ $\boxed{ab > n}$

But this is impossible, because $\boxed{a \cdot b = n}$

Successive Divisions

: dividing a number using smallest prime number. [This method is the step]

Prime Factorization

: Writing with only it's prime numbers product [The answer from successive division]

Ex: $36 : \frac{36}{2} \times \frac{18}{2} \times \frac{9}{3} \times \frac{3}{3}$ 1

$36 : 2^2 \times 3^2$

Ex: $7007 : \frac{7007}{7} \times \frac{1001}{7} \times \frac{143}{11} \frac{13}{13}$ 1

$7007 : 7^2 \times 11 \times 13$

GCD using Euclidean Algorithm

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gcd(a, b)
    if (a>b)
        a/b, remainder, change a with r
    else
        b/a, remainder, change b with r
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$\Rightarrow gcd(18, 48)$

$\Rightarrow \frac{48}{18} = 2 \Rightarrow 48-(18\times2) \Rightarrow r = 12$

$\Rightarrow gcd(18, 12)$

$\Rightarrow \frac{18}{12} = 1 \Rightarrow 18-(12\times1) \Rightarrow r = 6$

$\Rightarrow gcd(6, 12)$

$\Rightarrow \frac{12}{6} = 2 \Rightarrow 12-(6\times2) \Rightarrow r = 0$

So, $gcd(18, 48) = 6$

There are infinitely many primes

Using Euclid's theorem

$\therefore$ Suppose Q is the complete list of prime numbers

$$Q = P_1 \cdot P_2 \cdot P_3 \cdots P_n$$

$\because$ Now create a new number

$$N = (P_1 \cdot P_2 \cdot P_3 \cdots P_n) + 1$$

$\Rightarrow$ In this New number, N, $\frac{N}{P_1}$ leaves a remainder of 1.

$\Rightarrow$ But in our original list, Q, there was no remainder.

$\therefore$ That means:

- either N itself is a new prime,
- or N has a new prime factor.

[Proving the statement true]

Exponentiation by squaring / Binary Exponentiation

$\therefore 3^{645} \bmod 7$

$\Rightarrow (3^{322})^2 \cdot 3$

$\Rightarrow ((3^{161})^2)^2 \cdot 3$

$\Rightarrow (((3^{80})^2 \cdot 3)^2)^2 \cdot 3$

$\Rightarrow ((((3^{40})^2)^2 \cdot 3)^2)^2 \cdot 3$

$\Rightarrow (((((3^{20})^2)^2)^2 \cdot 3)^2)^2 \cdot 3$

$\Rightarrow ((((((3^{10})^2)^2)^2)^2 \cdot 3)^2)^2 \cdot 3$

$\Rightarrow (((((((3^{5})^2)^2)^2)^2)^2 \cdot 3)^2)^2 \cdot 3$

$\Rightarrow ((((((((3^{2})^2 \cdot 3)^2)^2)^2)^2)^2 \cdot 3)^2)^2 \cdot 3$

2 5 4 2 4 2 5 4 $\boxed{6}$

$\Rightarrow 9 \bmod 7 = 2$

$\Rightarrow 2^2 \times 3 = 12 \bmod 7 = 5$

$\Rightarrow 5^2 = 25 \bmod 7 = 4$

$\Rightarrow 4^2 = 16 \bmod 7 = 2$

$\Rightarrow 2^2 = 4 \bmod 7 = 4$

$\Rightarrow 4^2 = 16 \bmod 7 = 2$

$\Rightarrow 2^2 \times 3 = 12 \bmod 7 = 5$

$\Rightarrow 5^2 = 25 \bmod 7 = 4$

$\Rightarrow 4^2 \times 3 = 48 \bmod 7 = 6$

$\Rightarrow \boxed{3^{645} \bmod 7 = 6}$